Question and design

Does model width improve contact accuracy at the same one-epoch exposure? Baseline: exp277, native plus MPNN corpus, 24 layers. Hidden width 2048 -> 4096; MLP width 8192 -> 16384. Keep 32 query / 8 KV heads; head width 64 -> 128. Keep data, optimizer schedule, seeds and global batch fixed.

Architecture and smoke verified

5,861,984,256 parameters, versus exp277's 1,471,374,336 (3.984x). Architecture, tokenizer, data and recovery checks passed. 64-H100 smoke: ten updates, train loss 8.56948, eval loss 8.03575. 2.821 seconds/update; native and HF checkpoints saved with tokenizer. Compressed smoke schedule was spiky; this is an execution check.

Current status — October 26

Original stopped at user request at 12:40 EDT; driver and child confirmed killed. Last observed original step: 117,691, LR 0.001; numerically diverged. Fork continues on four nodes / 32 H100s at batch priority. Fork at cancellation check: step 116,485, LR 0.000707, train loss 2.82176. Original first training loss above 4 in the inspected window: step 77,229. Original validation rose from 3.09974 at 76,104 to 11.88947 at 78,218. It peaked at 42.43565 and remains 17.96501 at step 116,270. Only the lower-LR fork remains active; original is intentionally stopped. Fork final cooldown still starts at 213,076; total horizon 266,345 updates.

TPU trial succeeded

Interactive v5p-64: 32 chips across 8 VMs; all 20 updates completed. Finished October 16; no worker failures or preemptions. Native and HF step-19 checkpoints verified; tokenizer included. Median steps 5–19: 7.40 s/update, about 32% longer than GPU steps. Short native-only smoke; throughput is preliminary, not cost-normalized. Production continues on GPUs. No bulk transfer or migration performed.

Matched validation: lower LR gives a small advantage

The fork beats the default at all 19 exact-step post-fork evaluations. Same native + MPNN corpus, global batch, validation and absolute data position. At step 114,156: default 3.071639; fork 3.066603; original 17.649031. Fork minus default: -0.005036 nats/token, or 0.50% lower perplexity. This reverses the original wide model's disadvantage before the fork. Reducing LR is supported by stability and loss for this particular run. Single seed, incomplete epoch; width and LR differ from the smaller default. No contact-accuracy evaluation for the wider model yet.

Other architectures: context, not a common leaderboard

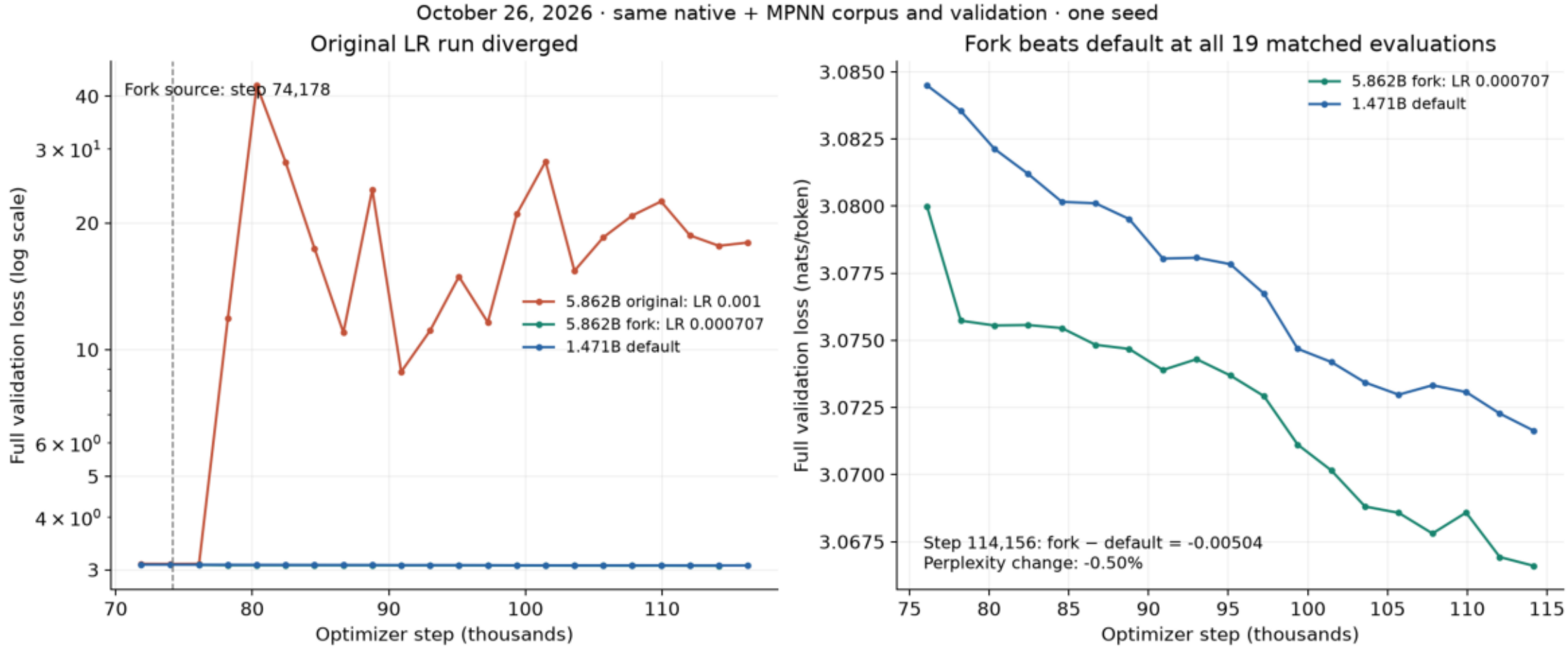
exp108 depth-scaled 2.93B: all three LR trials unstable around 6.8k updates. exp108 width-scaled 2.78B: all three trials finished; older recipe and validation. exp262 NoPE + smear: final loss 3.00207 versus its control's 2.97454. Its early advantage reversed, illustrating why interim rankings can mislead. exp157 learned RoPE deltas: corrected CE 3.0654 versus vanilla 3.0815. Only exp277 provides the same native + MPNN recipe as the current wider run. Raw historical validation losses cannot be pooled across cache/logging paths.

Parallel LR fork — October 22

Original schedule retains its 0.001 plateau and final cooldown. Fork restores checkpoint 74,178, resuming at absolute step 74,179. Weights, Adam moments/counters, RNG, data position and augmentation are preserved. Linear reduction to 0.000707 over 2,114 updates; target reached at step 76,293. Hold until original final cooldown at step 213,076; final LR target 0.0000707. Same 266,345-update horizon, separate W&B and checkpoint identities. Source copy and counters verified; 14 local configuration/restore/serialization tests passed. October 22 a02 failed before training; October 23 credentials refreshed. Retry a03 exposed CLI-class serialization; moved optimizer into an importable module. Corrected a04 is training; first nine LRs match the schedule within 1.14e-10. First loss at 74,179 exactly matches the original, confirming the starting comparison.

lr_fork_comparison.png

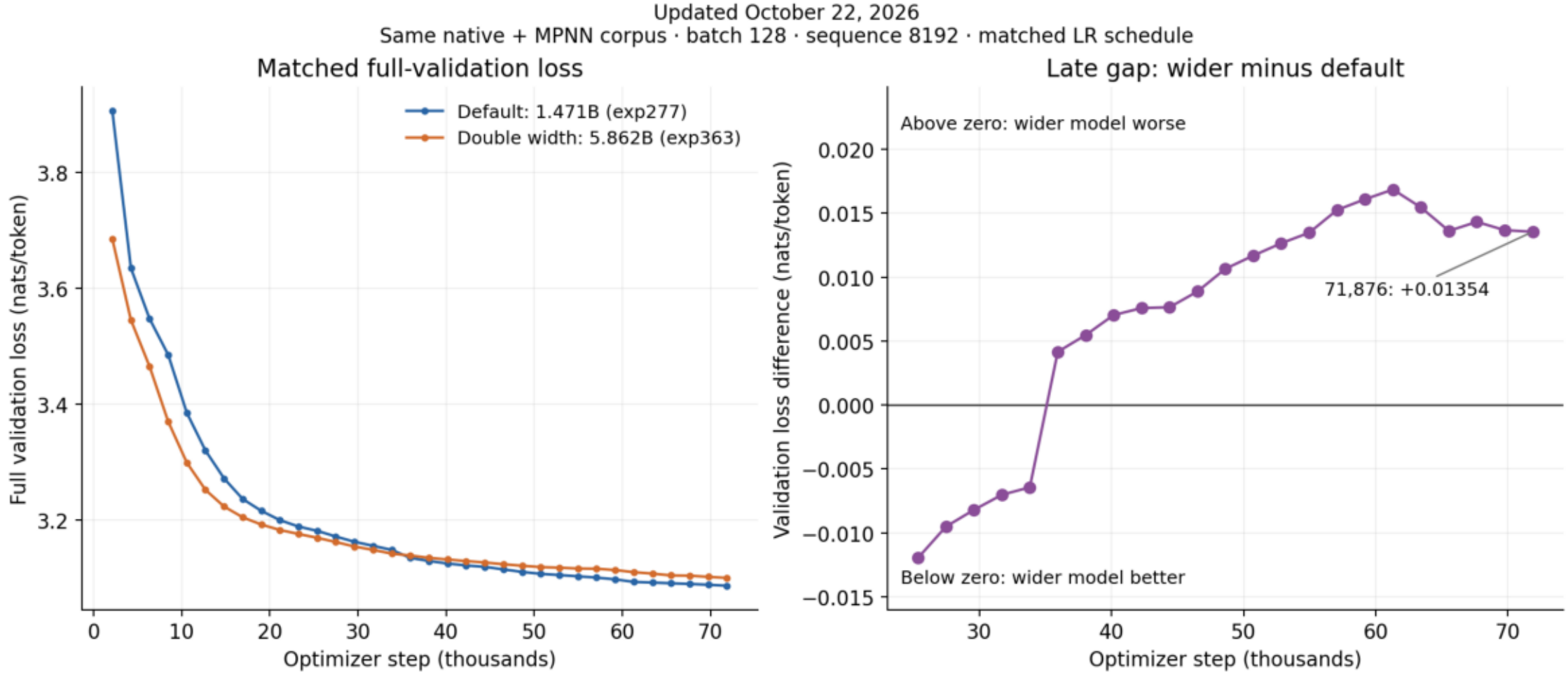
W&B snapshot 2026-10-26T15:55:10.854315+00:00. Original high-LR divergence is shown on a log scale at left; right compares fork/default at exact matching steps. No smoothing or interpolation. LR reaches 0.000707 at step 76,293. The small validation advantage does not establish contact accuracy.



$ plot_lr_fork.py

training_comparison.png

Full-validation observations through step 71,876, exported October 22, 2026. Exact-step comparison; no interpolation or smoothing. Same logical training recipe, different hardware/gradient accumulation; single seed, incomplete epoch. The right panel zooms into observations after step 25,000.




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$ plot_training_comparison.py
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